Degree-separated convolution and calibrated masks for a ten-integer consistency equation

Mathematical research prepared for ProveIt

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1 Degree-separated coding: a ten-integer equation

There is a completely fixed integer polynomial

$$F_{10}(f, g, h, k, l, w, x, y, z, m)$$

of exact total degree 8 469 528, evaluated by a division-free circuit of 533 binary arithmetic operations, for which ordinary mathematics formalizable in ZFC proves

$$\neg\exists (f, g, h, k, l, w, x, y, z, m) \in \mathbb{Z}^{10}\ (F_{10} = 0) \quad \Longleftrightarrow \quad \mathrm{Con(ZFC)}.$$

This is a new source-coding architecture, not a new unknown-count theorem. The inherited compiler calibration and Sun arithmetic bridge remain stated mathematical dependencies. No consistency decision, global minimality claim, or new end-to-end proof-assistant certificate is asserted.

The structural change is to split the extraction convolution by homogeneous degree. A linear label may now collide with a quadratic label: their kernels are separate summands. This relaxes the finite isolation problem rather than merely finding a better placement for the previous problem. Elementary row operations, one shared positivity shift, borrow-tolerant target padding, and exact mask-population calibration complete the construction.

1.1 Fixed source and three homogeneous kernels

Use exactly the unshifted, strictly positive, 31-coordinate parent in `data/parent_quartic31.json`, with its complete original four-port binding $(\beta_{\mathrm{native}}, K_0, M_C, M_F)$. The four included compressed payloads have bit lengths $(1064921, 1070004, 1064921, 1064922)$; the last is `MF_source`, already shifted, and is not shifted again. `INSTANCE.json` is the active binding. No package-52 transport row or coefficient is spliced into this tuple. The pinned parent theorem is

$$\exists p \in \mathbb{Z}_{>0}^{31}\ r_1(p) = \cdots = r_{22}(p) = 0 \quad \Longleftrightarrow \quad \neg\,\mathrm{Con(ZFC)}.$$

Replace only r_{14} by $\rho_{14} = r_{14} + r_2$ and r_{15} by $\rho_{15} = r_{15} + r_3$; otherwise put $\rho_j = r_j$. The inverse row operations preserve the common zero set over every coefficient tuple. There are now 89 records: seven constant, 39 linear, and 43 quadratic.

In the parent coordinate order

$$(J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma,\\ N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B_{\mathrm{aux}}, \Delta),$$

choose the positive, distinct weights

$$(w_1, \ldots, w_{31}) = (1, 7, 61, 36, 16, 85, 20, 67, 43, 11, 34, 13, 96, 80, 29, 50,\\ 2, 109, 110, 112, 116, 91, 53, 55, 27, 101, 92, 9, 107, 37, 60).$$

The row order and targets are

$$(2, 22, 8, 15, 9, 11, 3, 17, 13, 18, 12, 20, 4, 21, 19, 1, 6, 10, 16, 14, 7, 5),$$
$$(28, 85, 286, 384, 593, 602, 642, 731, 906, 966, 1083, 1212, 1231, 1393,$$
$$1461, 1471, 1678, 1719, 1732, 1880, 2107, 2129).$$

Let q_j denote the target assigned to row j, and write $\rho_j = \sum_\alpha a_{j,\alpha} p^\alpha$. For $d = |\alpha|$ put

$$e(\alpha) = \sum_i \alpha_i w_i, \qquad c_{j,\alpha} = \begin{cases} 2a_{j,\alpha}, & d = 0, 1, \\ a_{j,\alpha} \prod_i \alpha_i!, & d = 2. \end{cases}$$

Define the fixed integer polynomials

$$D_d(Z) = \sum_{j,\,|\alpha|=d} c_{j,\alpha} Z^{q_j - e(\alpha)}, \qquad U_p(Z) = \sum_i p_i Z^{w_i}, \qquad G_p(Z) = D_2(Z) U_p(Z)^2 + D_1(Z) U_p(Z) + D_0(Z).$$

All reversed exponents are nonnegative. The exact finite certificate checks, for every used degree-d monomial α and every degree-d monomial γ in 31 variables,

$$q_j - e(\alpha) + e(\gamma) = q_k \quad \Longleftrightarrow \quad j = k \text{ and } \alpha = \gamma.$$

There are respectively $1, 31, 496$ expansion monomials in degrees $0, 1, 2$, and 22,544 source/expansion interactions. There is no requirement to separate a degree-one label from a degree-two label. Multinomial expansion and the displayed coefficient weights therefore give the universal identities

$$[Z^{q_j}] G_p(Z) = 2\rho_j(p) \quad (1 \le j \le 22). \tag{1}$$

The independent checker compares the entire extracted polynomials in all 31 coordinates and four ports, not just evaluations at test points.

Write $M = 116$, $Q = 2129$. The exact kernel degrees and convolution height are

$$\deg D_0 = 1719, \qquad \deg D_1 = 2027, \qquad \deg D_2 = 1929, \qquad H = \max_d (\deg D_d + dM) = 2161.$$

The leading term of D_2 is exactly $+Z^{1929}$, arising from EX in row five. Every kernel separately has distinct exponents. Replacing each port monomial in a coefficient by its strict bit bound and taking absolute integer coefficients gives a maximum majorant of bit length 2134927. Consequently, with

$$C_0 = 2^{2134927}, \qquad \kappa = 2134940, \qquad \beta_S = 2^\kappa,$$

every kernel coefficient has absolute value less than C_0, and $\beta_S > 4096 C_0$.

1.2 Source polynomials and pre-binary bounds

In the unrestricted integer unknowns f, g define

$$b = 1 + 3f, \quad B = \beta_S b^2, \quad \ell = \sum_i B^{w_i}, \quad U_c = \ell + g,$$
$$G_c = D_2(B) U_c^2 + D_1(B) U_c + D_0(B), \quad A_T = \sum_j B^{q_j},$$
$$N_0 = B^{117}, \quad N_1 = B^{2162}, \quad N = N_0 N_1 = B^{2279},$$
$$M_0 = N_0 - 1 - (b-1)\ell, \quad J_c = 2G_c + A_T,$$
$$S_c = g + J_c N_0, \quad T_c = M_0 + (B-2) A_T N_0,$$
$$X = N^2 + (S_c + T_c) N + N - 1 - S_c,$$

$$Y = \beta_S^{15}\, 2^{\kappa-21} B^{2402} b = 2^{2418\kappa-21} b^{4805}, \qquad \mathcal{C} = bB^{116} = 2^{116\kappa} b^{233}.$$

All powers have fixed nonnegative integer exponents. Every displayed register is an abbreviation, never an additional quantified unknown.

Assume $f \neq 0$, b is a square, and $0 \leq g < 2\mathcal{C}$. Then $b \equiv 1 \pmod 3$ implies $b \geq 4$, and $B > 4096 C_0 b^2$. Distinct positive weights give $\ell < 2B^{116}$, hence

$$B^{116} \leq U_c < 3bB^{116}.$$

In particular, without assuming b is a power of two,

$$D_2(B) > \tfrac{1}{2} B^{1929}, \quad |D_1(B)| < 2C_0 B^{2027}, \quad |D_0(B)| < 2C_0 B^{1719}.$$

After division by $B^{1929} U_c^2$, the two possible negative terms are bounded by $2C_0/B^{18} < 1/8$ and $2C_0/B^{442} < 1/8$. Thus $G_c > \frac{1}{4} B^{1929} U_c^2 > 0$. This is the necessary signed leading-layer argument; positivity is not inferred from positive inputs alone.

The kernel degree bounds also give

$$|G_c| < 2C_0(1 + 3b + 9b^2) B^H \leq 26 C_0 b^2 B^H, \qquad 2G_c < N_1/64.$$

Since $Q = 2129 < H$, both A_T and $(B-2)A_T$ are less than $N_1/64$. It follows that

$$0 < J_c < N_1/32, \qquad J_c + (B-2)A_T + 2 < N_1.$$

The digits of M_0 are $B-1$, except that the 31 designated cells have digit $B-b$. Therefore $0 \leq M_0 < N_0$, and $g < N_0$ as well. The preceding strict high-block bound proves

$$0 \leq S_c, \quad 0 \leq T_c, \qquad S_c + T_c < N. \tag{2}$$

Consequently $X \geq N^2 > 3b$ and $Y > \max(b, 256)$. Because κ is even, $B \equiv N \equiv 1 \pmod 3$, $M_0 \equiv 0$, and $T_c \equiv -22$; hence $X \equiv 1 + T_c \equiv 0 \pmod 3$. These are all established before binary-base recovery.

1.3 Borrow-tolerant extraction and exact mask population

Suppose now that b is a power of two. So are B, N_0, N. The low mask condition $g \,\&\, M_0 = 0$ is exactly

$$g = \sum_i n_i B^{w_i}, \quad 0 \leq n_i < b.$$

For these digits put $p_i = n_i + 1$, so $U_c = \sum_i p_i B^{w_i}$ encodes positive parent coordinates without expanding a successor substitution in every residual. Write $h_s = [Z^s] G_p(Z)$ and let ϵ_s be the indicator of the 22 targets. All coefficients, including unsupported collisions, satisfy

$$|h_s| < C_0\{(\sum_i p_i)^2 + \sum_i p_i + 1\} \leq 993 C_0 b^2 < B/4.$$

The signed expansion of J_c is $\sum_{s=0}^{H} (2h_s + \epsilon_s) B^s$. Euclidean digit normalization has incoming carries only in $\{-1, 0\}$:

$$u_0 = 0, \quad u_{s+1} = \left\lfloor \frac{2h_s + \epsilon_s + u_s}{B} \right\rfloor, \quad d_s = 2h_s + \epsilon_s + u_s - B u_{s+1}.$$

At a marked cell the mask $B-2$ admits precisely digits $0, 1$. If $h_s = 0$, the unnormalized digit $1 + u_s$ passes. If $h_s \geq 1$, it lies in $[2, B)$ and fails. If $h_s \leq -1$, it lies strictly between $-B+1$ and zero, and its normalized digit is at least two, so it fails. Thus incoming borrows do not affect the zero test. Positivity of J_c was proved independently, ensuring these are its ordinary digits. Combining the disjoint low and high blocks with (1) proves

$$S_c \,\&\, T_c = 0 \quad \Longleftrightarrow \quad g = \sum_i n_i B^{w_i},\ 0 \leq n_i < b, \quad r_1(n+1) = \cdots = r_{22}(n+1) = 0.$$

For $N = 2^u$ and $S, T \geq 0$ with $S + T < N$, let $X_* = N^2 + (S+T)N + N - 1 - S$. Its disjoint binary blocks give

$$s_2(X_*) = 1 + u + s_2(T) - \{s_2(S) + s_2(T) - s_2(S+T)\}.$$

The braces count carries with multiplicity, and vanish precisely when $S \,\&\, T = 0$. Legendre's formula therefore implies

$$2^{1+u+s_2(T)} \mid \binom{2X_*}{X_*} \quad \Longleftrightarrow \quad S \,\&\, T = 0.$$

The stronger range (2) is essential. Writing $b = 2^k$, $B = 2^v$, the actual mask population is

$$s_2(T_c) = (117 + 22)v - 31k - 22, \qquad u = 2279v.$$

Hence $1 + u + s_2(T_c) = 2418v - 31k - 21 = \log_2 Y$, exactly explaining the fixed polynomial threshold used above.

Conversely, a positive parent zero is not the all-one vector: original row five equals one there. For arbitrarily large $b = 4^s$ exceeding all its coordinates, encode $n_i = p_i - 1$. Then $g \geq B > b$ and $g < \mathcal{C}$; the latter follows from $(b-1)\ell < bB^{116}$. All marked tests pass and $Y \mid \binom{2X}{X}$. This proves both directions of the entire source contract needed by the inherited finalizer, including its lower bound on forward witnesses and its pre-binary reverse bounds.

1.4 Explicit radical-free finalizer

The following formulas are the inherited package-31 quintic, merged-square, structural-weight finalizer. They are written out to fix the actual equation. Distinguish its local B_*, S_*, T_* from the source base and masks above. Define

$$\begin{gathered}
L = lY, \qquad U = 2LX, \qquad V = 4gwY, \qquad W = bw, \\
A = U(V+1), \qquad B_* = 2X + 1, \qquad C = B_* + (A-2)h, \\
P = U^2 V, \qquad K = X + 1 + k(P-2), \\
S_* = 2A - 5, \qquad T_* = 3WC - 2(W^2 - 1), \\
R_2 = (P^2 - 4)K^2 + 4, \\
\Delta = 8\mathcal{C}^3 g K^2 - g^2\{32(C - KL)^2\mathcal{C}^3 + g^2 K^2\}, \\
O = f^2 l^2 \Delta, \qquad \xi = zx, \\
D = (A^2 - 4)C^2 + 4, \qquad E = C^2 D\xi, \\
F = 4(A^2 - 4)E^2 + 1, \qquad G_* = 2(A+2)(A-2)^2 E^2 - 1, \\
\Omega = 16E, \qquad H_* = C + (B_* - C)F + \Omega F y, \\
I_* = (G_*^2 - 1)H_*^2 + 1.
\end{gathered}$$

Put

$$\begin{aligned}
r &= T_* + S_* y, & \Sigma &= \xi S_*^5, & & \\
\Theta &= (r^3 - rS_*^2 - S_*^3)(r^2 + 23S_*^2), & & & & \\
a_0 &= b\{1 + (3O - 4)b^2 z^2\}, & a_1 &= R_2, & a_2 &= DFI_*, \\
w_0 &= 1, & w_1 &= a_0, & w_2 &= a_0(P^2 + K^2 + 3), \\
Z &= \Sigma m - \Theta, & A_i &= \Sigma^2 w_i^2 a_i \quad (0 \leq i \leq 2), & & \\
V_n &= Z^2 + A_2 - A_0 - A_1, & W_n &= V_n^2 + 4Z^2 A_2 - 4A_0 A_1. & &
\end{aligned}$$

The promised single equation is

$$\boxed{F_{10} = W_n^2 - 16 V_n^2 Z^2 A_2 = 0.} \tag{3}$$

Every expression is an integer polynomial. In particular, radicals and binomial coefficients appear in the proof only, never in the equation.

Exact dependency boundary. Sun's 2021 Theorems 4.12–4.13, specialized as in his 2026 Propositions 6.1–6.2, provide the two bridge directions for any source satisfying the contract just proved. Forward witnesses have b square, R_2 square, $S_* \mid T_*$, $C = u_{B_*}(A)$ and $O > 0$. Reverse witnesses with b square, $O > 0$, $\xi \neq 0$, R_2, DFI_* square and $S_* \mid T_*$ recover $b \in 2\uparrow$, $0 < g < 2\mathcal{C}$ and the required binomial divisibility. The residue interpolation theorem permits ξ to be divisible by any prescribed positive integer and permits any residue class for $y \pmod{\xi}$. The only data substituted here are the newly proved source polynomials; the bridge does not require $Y = N^2$.

For completeness, the inherited finalizer implications can be verified as follows. The polynomial $(v^3 - v - 1)(v^2 + 23)$ has a root modulo every positive integer and none over $\mathbb{Q}$. For odd primes other than 23 the cubic discriminant is -23: if the cubic is irreducible its Frobenius permutation is even, so the quadratic has a simple root; otherwise the cubic has a simple root. At 23 the cubic has simple root 3. At 2, $-23 \equiv 1 \pmod 8$. Hensel lifting and the Chinese remainder theorem finish the local-root argument. Prime valuations in the two homogeneous factors show that $\Sigma \mid \Theta$ forces $S_* \mid T_*$: if $v_p(S_*) > v_p(T_*)$, the valuation of Θ is $5v_p(T_*) < 5v_p(S_*)$. Since S_* is odd, it never vanishes; the absence of rational roots ensures $\Theta \neq 0$.

When $z \neq 0$, a_0 square implies that its coprime factors b and $1 + (3O - 4)b^2z^2$ are signed squares. The congruence $b \equiv 1 \pmod 3$ excludes a negative square and excludes zero, including the zero-factor exception. Thus b is a positive square and the usual positivity Pell lemma gives $O > 0$. Conversely, forward witnesses have $O \geq 4$: the positive integer Δ is divisible by $g \geq b \geq 4$. For $d = 3O - 4 > 0$, $d \equiv 2 \pmod 3$ is nonsquare. A positive Pell unit raised to its order modulo b gives $v = bz > 0$ with $u^2 - dv^2 = 1$, making a_0 a square. Residue interpolation then supplies $\xi = zx$ and a y that simultaneously makes a_2 square and $\Sigma \mid \Theta$.

If $z \neq 0$, the displayed a_0 always has $|a_0| > 1$. The exceptional possibilities $a_0 = 0, \pm 1$ force $|b| = 1$, hence $b = 1$, $f = O = 0$, and $a_0 = 1 - 4z^2$, a contradiction. Also $\sqrt{|a_1|} \leq P^2 + K^2 + 2$. Thus the weights obey

$$|w_1| > \sqrt{|a_0|}, \qquad |w_2| > \sqrt{|a_0|} + |w_1|\sqrt{|a_1|}.$$

In the multiquadratic field, an automorphism changing a nonzero weighted root has a largest changed index; its magnitude exceeds the sum of all smaller changed magnitudes. A rational weighted sum can therefore occur only when every a_i is an integer square. This also covers negative radicands by complex absolute values, and zero radicands separately.

The exact norm identity

$$F_{10} = \prod_{\epsilon \in \{\pm 1\}^3} \left(Z + \sum_{i=0}^{2} \epsilon_i \sqrt{A_i} \right)$$

has 35 expanded terms and is checked independently by integer polynomial arithmetic. If $\Sigma = 0$ then $F_{10} = \Theta^8 \neq 0$. At a zero of F_{10} we therefore have $\Sigma \neq 0$, hence $z \neq 0$; separation forces all a_i square and $\Sigma \mid \Theta$. The reverse bridge and source proof recover a parent solution. Conversely the forward construction just described makes all a_i square and $\Sigma \mid \Theta$; choosing m to cancel one signed factor gives $F_{10} = 0$. Composition with the pinned parent theorem proves the stated consistency equivalence in ZFC.

1.5 Exact circuit, degree, and reproducibility

The actual circuit uses the same finalizer polynomial in factored form. For example, with $A = 2A_{1/2}$ and $F - 1 = 4(A^2 - 4)E^2$, its Lucas expression is $G_* = (A_{1/2} - 1)(F - 1) - 1$; also $H_* = C + F(B_* - C + \Omega y)$. The common factor $\Sigma^2 a_0^2$ is reused in the last two scaled radicands. Fixed powers reuse already computed powers B^i, B^j when $i + j$ is the required exponent, falling back to binary exponentiation otherwise. These are exact polynomial identities, not a relaxation of the equation.

`generated/sun10.json` contains all ten variable leaves and 533 live binary gates: 340 multiplications, 166 additions, and 27 subtractions. Multiplication by a numeral is charged. The fixed coefficient leaves are

specified by `INSTANCE.json`, the four payloads, and `generated/source_records.json`. Their construction is not included in this arithmetic-operation count, which is not bit complexity.

The source degrees are $\deg X = 9116$, $\deg Y = 4805$, and $\deg \mathcal{C} = 233$. The propagated final upper bound is

$$456(9116) + 896(4805) + 24(233) + 1760 = 8\,469\,528.$$

After all ten variables are replaced by one indeterminate T, the coefficient of $T^{8469528}$ has nonzero residues

p	101	1009	1000003
$[T^{8469528}]F_{10}(T,\dots,T) \bmod p$	68	256	476348.

Any one residue proves the matching lower bound, so this degree is exact. This is leading-coefficient propagation, not a random point evaluation.

The independent checker imports no builder. It independently parses all 22 source residuals; verifies the inverse row operations, 89 records, and all 22,544 isolation interactions; checks 59 local circuit identities, the additional Lucas factorization, and the eight-factor norm identity; audits every gate and fixed binding; and certifies the exact degree. Finite implementation tests cover 476 borrow transitions, 11,049 binary block pairs, and 60 large-integer source assignments, including nonbinary square bases 25 and 49. These finite checks supplement, rather than replace, the unbounded mathematical arguments. Both ordinary Python and `python3 -O independent_check.py` pass.

Against the prior calibrated-mask ten-integer construction of degree 21 111 400 and 659 gates, the new point has 59.88% lower degree and 126 fewer gates (19.12%). Against the incoming 582-gate homogeneous source point of degree 50 822 936, it has 49 fewer gates and more than sixfold lower degree. These are explicit pinned comparisons, not claims of global optimality or superiority over every variable/domain tradeoff.

The inherited exact-source pin is `124c07c0b77a9e20e84676514d8b1ff78536ea3f`; `SOURCE_PINS.json` preserves all 12 source identities. Primary arithmetic dependencies are Z.-W. Sun, *Further results on Hilbert's Tenth Problem*, Sci. China Math. 64 (2021), 281–306, https://arxiv.org/html/1704.03504v7; and *Ten unknowns for Hilbert's tenth problem over the integers*, 2026 preprint, https://arxiv.org/html/2609.01594v2. The complete inherited source, coefficient, and finalizer proof records are preserved under `upstream/`; their historical relative paths do not supersede the active manifest.